

CEN 4010 Team Project – Iteration 2

Implemented Requirements:

The team has continued to update the code by adding more of our requirements and refactoring code, so it stays consistent and working. We decided to change the UI to make it less compact and implement hiding elements, so the screen is not so visually cluttered. With the free space that the UI change provided us, we decided to implement the possibility of a sheet of music to be two lines instead of just one. We wanted the user to be able to return to the instruction screen, so we implemented a way to go backwards through the screens. Lastly, we added support for multiple different types of notes and rests to move closer to our vision for the final product. These are all the implemented requirements that we decided to work on for Iteration 2:

- When the user presses ENTER when on the start screen, the instruction screen displays and an image of the keyboard inputs is displayed.
- The system reads MIDI controller inputs in addition to keyboard ones for the input of the notes.
- After the user completes an attempt, accuracy scores get displayed, and when they press ENTER to start a new attempt, the accuracy scores hide.
- Sheet music consists of more than only quarter notes. May contain whole, half, eighth, or sixteenth notes. May also contain accidentals.
- After the user enters BACKSPACE, the previous screen gets displayed unless on the start screen.
- On the practice screen, the size of the measures varies depending on the number of notes/rests in each measure.
- When a sheet of music is generated, two lines of notes are displayed if there are an adequate number of notes/rests in a measure.

After iteration 1, we decided to complete most of the necessary tasks to allot time for the big requirements of Iteration 3 which are creating the algorithm so the music sheet will be reasonable and random, and calculating the rhythm accuracy. Not all these requirements were fully fleshed out as we discovered some areas of improvement while implementing Iteration 2; as an example, we improved the system by implementing multiple lines of sheet music. Our system is nearing completion, and the finished shell is visible. We will continue to improve our code and solutions in Iteration 3.

Our Junit test cases mainly focus on the notes, rests, measures and staff. They are particularly focused on if they are being created and manipulated in the ideal way. For example, with the Duration enumeration it is checked whether the right time units have been applied or not. Additionally, the Measure class is tested by checking if notes and rests are correctly being added to a newly created measure. One last example includes

the Note class where multiple pitch shifts are calculated, then checked if these match the calculations done in the class.

Use Cases:

The team identified the following use cases based on the requirements we chose to implement for this iteration:

1. **Name:** Navigate Screens

Actor: User

Preconditions: None (program just opened)

Description: The user just opened the program and is met with the start page. They then navigate to the practice screen but decide to check the instruction screen once more before returning to the practice screen.

Steps (Two-column Format):

Actor Actions	System Responses
1. User presses ENTER key	2. Instructions screen is displayed
3. User presses ENTER key	4. Practice screen is displayed
5. User presses BACKSPACE key	6. Instruction screen is displayed
7. User pressed ENTER key	8. Practice screen is displayed

2. **Name:** Input Keys

Actor: User

Preconditions: User is at the practice screen

Description: The user is at the practice screen and presses note inputs related to the notes they see on the screen. The system reads each input and colors the note gray if correct and pink if incorrect.

Steps (Two-column Format):

Actor Actions	System Responses
1. User enters correct musical note input	2. Current note is colored gray
3. User enters incorrect musical note input	4. Current note is colored pink
5. User repeats step 1 and 3 until the entered notes equals the number of notes on the staff	6. System stops accepting note input until a new attempt is started.

3. **Name:** View Pitch Accuracy Score

Actor: User

Related Use Cases: Includes: Input keys

Preconditions: User is at the practice screen

Description: The user enters note input (through the keyboard for this iteration), attempting to match the sequence of notes on the staff. Finally, the system displays a pitch accuracy score upon the user entering a number of notes that equals the number of notes on the staff.

Steps (Two-column Format):

Actor Actions	System Responses
1. Input keys	2. Attempt counter is incremented 3. Pitch accuracy score is displayed

4. **Name:** Reset Attempt Counter

Actor: User

Related Use Cases: Includes: Input keys

Preconditions: User is at the practice screen and has a number of current attempts

Description: The user enters note input, attempting to match the sequence of notes on the staff. The user is displayed a pitch accuracy score upon entering a number of notes that equals the number of notes on the staff. After some number of attempts as described above, the user makes the optional choice to reset the attempt counter, which sets the attempt counter and average pitch score to 0 (default when no attempts have been completed).

Steps (Two-column Format):

Actor Actions	System Responses
1. Input keys	2. Attempt counter is incremented 3. Pitch accuracy score is displayed
4. User enters CTRL + R	5. Attempt counter is set to 0 6. Average pitch accuracy between attempts is set to 0

5. **Name:** Start New Attempt

Actor: User

Related Use Cases: Includes: Input keys

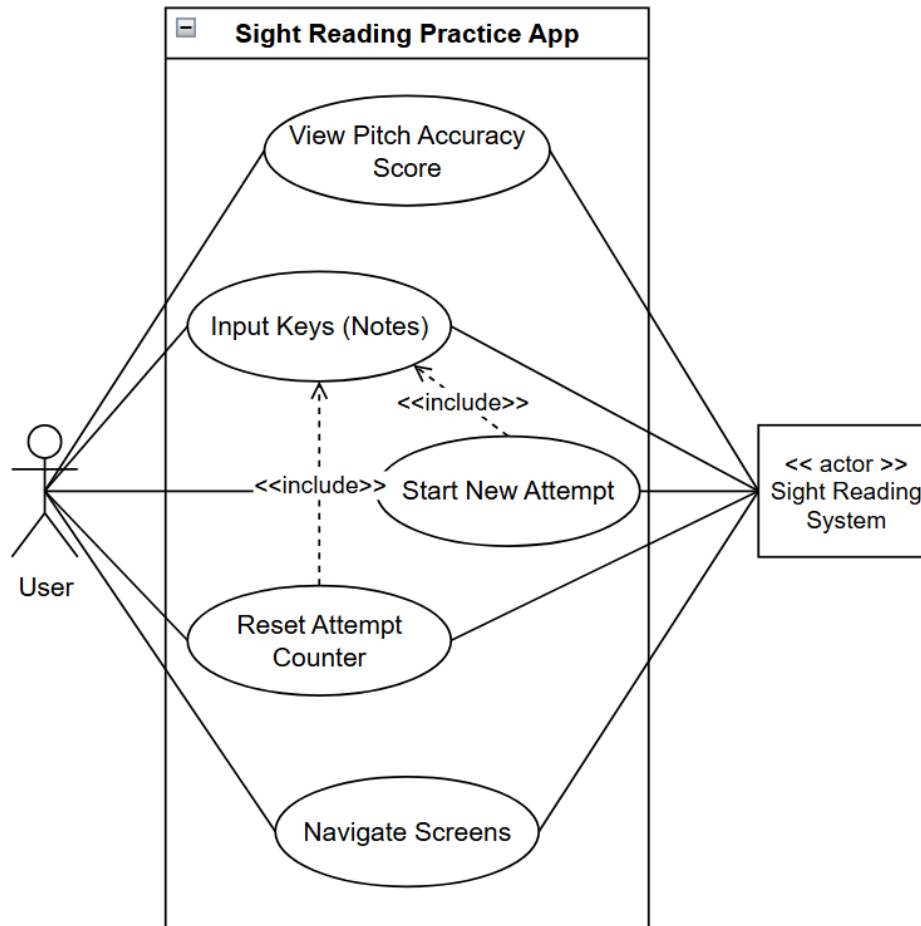
Preconditions: User is at the practice screen

Description: The user enters note input, attempting to match the sequence of notes on the staff. The user is displayed a pitch accuracy score upon entering a number of notes that equals the number of notes on the staff. The accuracy scores get displayed and the user continues to the next attempt

Steps (Two-column Format):

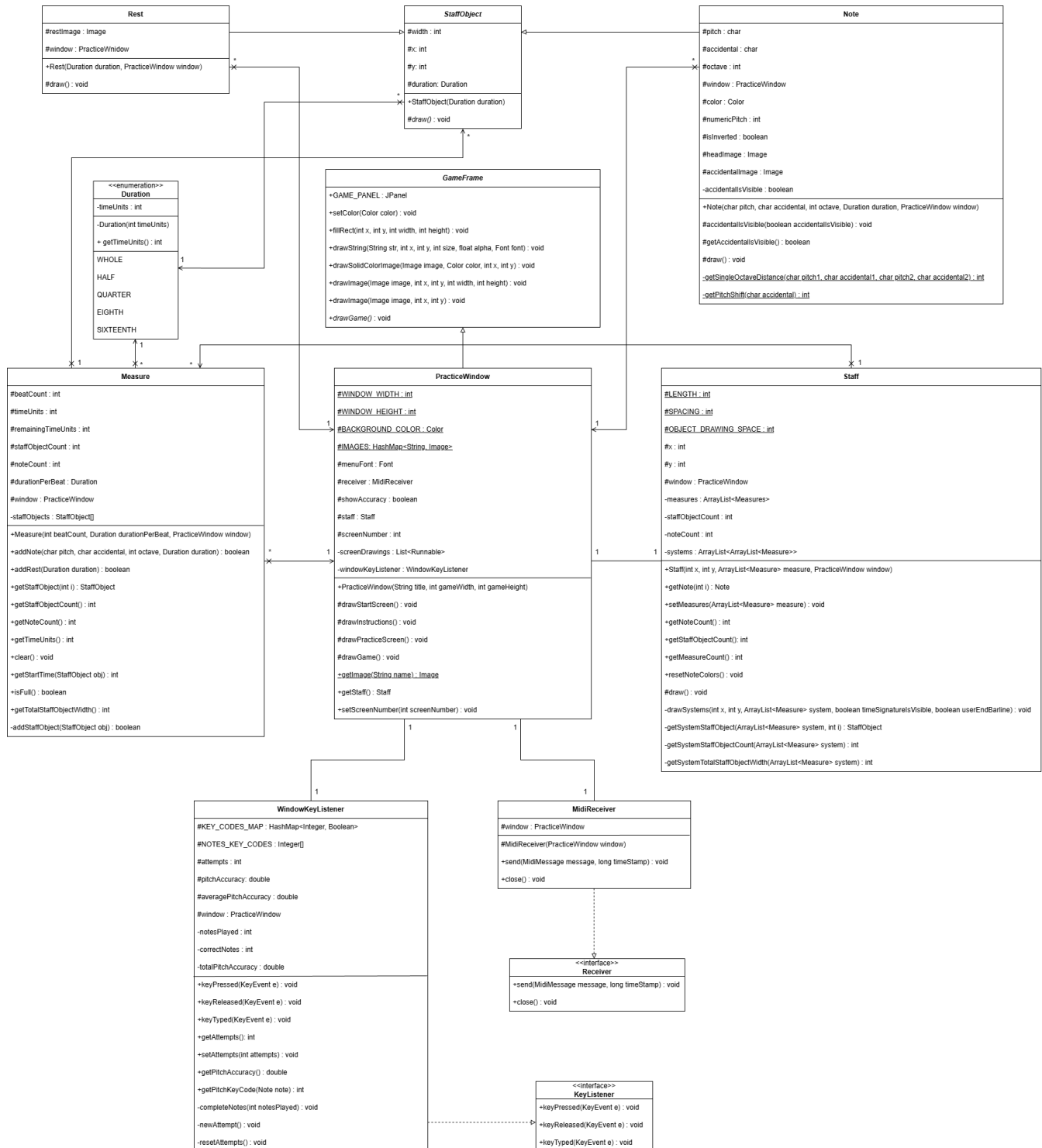
Actor Actions	System Responses
1. Input keys	2. Attempt counter is incremented 3. Pitch accuracy score is displayed
4. User presses ENTER key	5. Accuracy scores are hidden 6. The colored notes turn black

Use Case Diagram:



We updated the use case diagram in accordance with our implementations for Iteration 2. That includes adding the actor for the system and creating use cases for Navigating Screens and Inputting Keys. Inputting Keys has sub-test cases since it is included in Start New Attempt and Reset Attempt Counter.

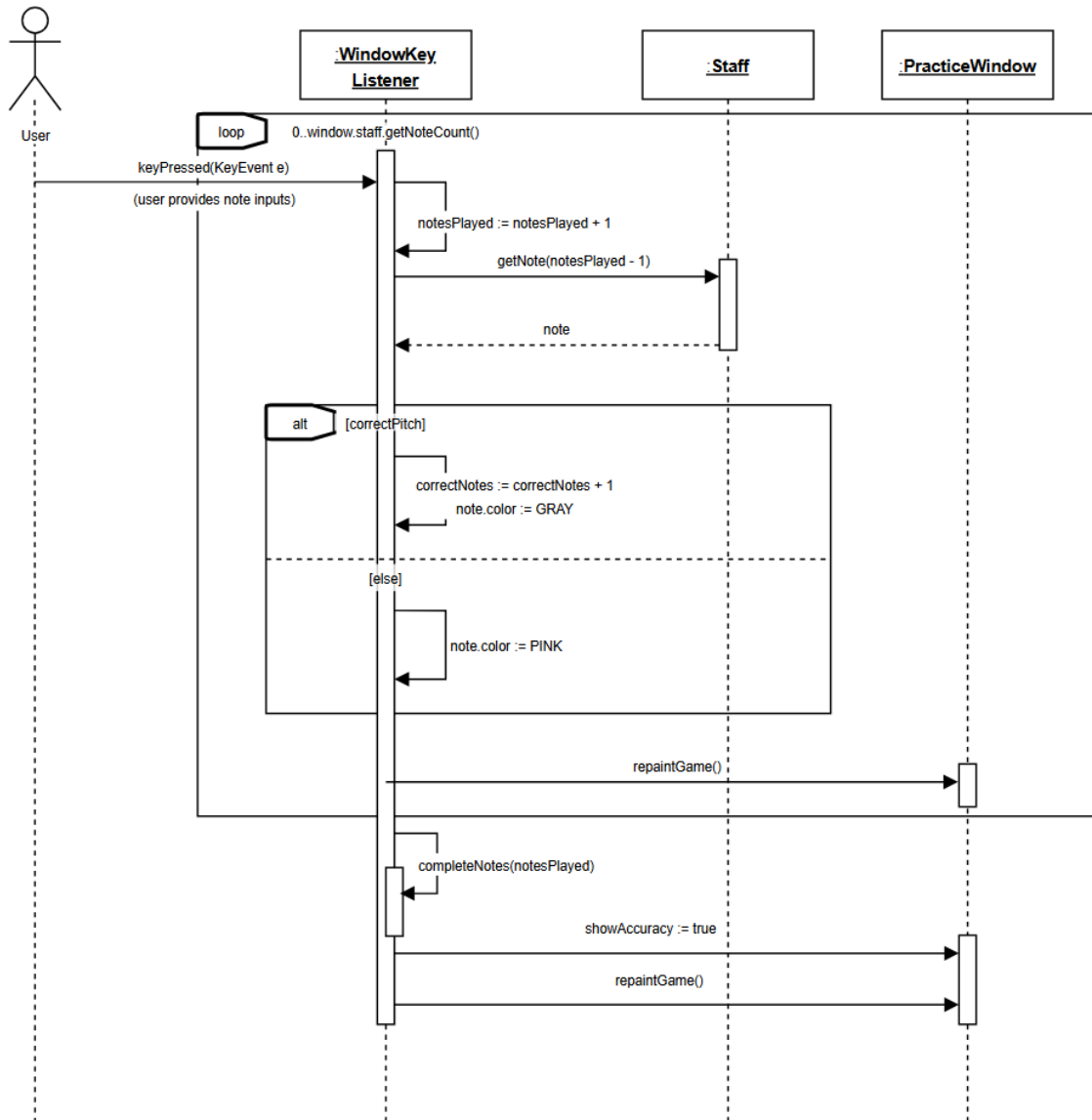
UML Diagram:



The new UML diagram captures our new design decisions made during this iteration, which involved the creation of many new classes and introduced new class relationships as seen in the diagram.

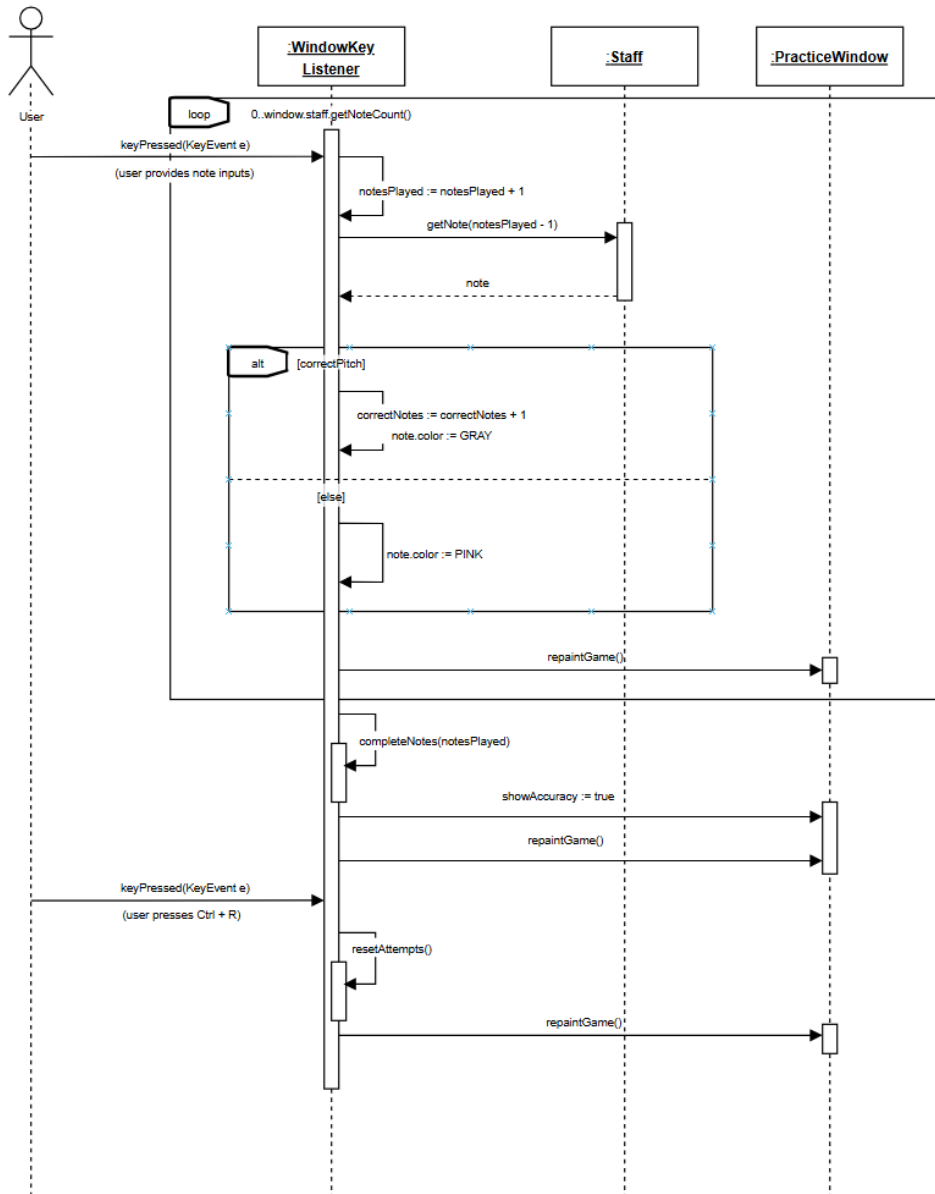
Sequence Diagrams:

1. View Pitch Accuracy Score



This is a sequence diagram for Get Pitch Accuracy. The objects we have are WindowKeyListener, Staff, and PracticeWindow. The outer loop signifies the KeyListener constantly waiting for input (notes) until the notes inputted is equal to or greater than the number of notes on the current sheet of music. The alternative section colors the note GRAY if correctPitch is true. Otherwise, it colors the note PINK. The game is then repainted, and the accuracy scores are displayed.

2. Reset Attempt Counter



This is a sequence diagram for Reset Attempt Counter. The user goes through one attempt by entering notes until the number of notes is equal to or greater than the notes on the music sheet. The user sees their accuracy score and then presses CTRL + R to reset the attempt counter.

Poster:

Sight-Reading Practice Application

AUTHORS

Christian Rios, Kemuel Dorset, and Anthony Noschese

COURSE

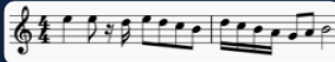
CEN 4010: Software Engineering
Instructor: Dr. Upulee Kanewala



INTRODUCTION

Our application aims to solve a problem for learners of the piano. Sight-reading is the act of playing an instrument according to sheet music that the player has never seen or practiced beforehand.

For example, a piano learner might want to sight-read the following passage or similar passages:



An example passage to be sight-read by a beginner pianist.

Without a teacher, it can be difficult for beginner pianists to learn how to sight-read. Learning how to sight-read is important for pianists as it allows them to learn musical pieces faster and play music without memorizing. Really, sight-reading is an important component of musical proficiency for any instrument player. However, not all pianists have access to music teachers to help them develop the skill and provide the feedback they need to improve. This problem of feedback is what our application aims to solve.

As a black box, our solution starts with generating random lines of sheet music (realistic to what a pianist might see in practice) and detecting inputs from a digital piano connected to a computer. As a pianist tries to play according to the sheet music, our software gives visual feedback by changing the colors of notes on the page. After the pianist finishes playing the passage, accuracy scores based on rhythm and pitch are displayed for additional feedback.

DESIGN

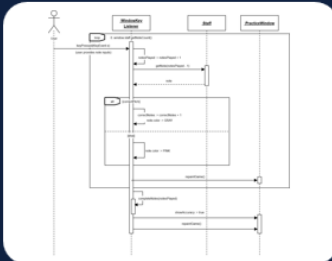
The following diagrams highlight interesting parts of our software's design (which we may explain further).



Use case diagram for the project.



UML class diagram section showing the design of the staff.



Sequence diagram for the "View Pitch Accuracy Score" use case.

Our use case diagram helps us identify the fundamental user actions we want our program to support and see the relationships between these actions. We will continue to update it as we finish the project.

TECHNOLOGIES

Our application was built using the Java programming language using classes from Java's Abstract Window Toolkit (AWT) and Swing libraries as the basis for the application's GUI.

The application reads data from a connected digital piano using Java's MIDI library. MIDI stands for Musical Instrument Digital Interface and is the standard protocol for reading inputs from digital instruments connected to computers. For our program, it importantly allows us to detect when notes are played and the pitches of those notes.

NEXT STEPS

To finish our course project, we plan to finish implementing the following features:

- Algorithm for random sheet music generation
- Rhythm accuracy calculation
- Beaming of shorter note durations within beats

However, there are many features that should be implemented in the future (that are beyond the scope of our course project) to make the system more useful. The following are a few of these features:

- More sheet music features (dots, ties, key signatures, grand staff, etc.).
- Customization options (choose key/time signature).
- Difficulty level option for generated passages.

(Separate PDF file included)

Commit Logs:

Kemuel's Logs:

6cf1756 refs/heads/Kemuel@{0}: pull --no-stat -v --progress origin master: Merge made by the 'ort' strategy.

b963299 refs/remotes/origin/master@{0}: pull --no-stat -v --progress origin master: fast-forward

6cf1756 HEAD@{0}: pull --no-stat -v --progress origin master: Merge made by the 'ort' strategy.

dd84e73 refs/remotes/origin/Kemuel@{0}: update by push

dd84e73 refs/heads/Kemuel@{1}: commit: Corrected use case diagram

dd84e73 HEAD@{1}: commit: Corrected use case diagram

b41c579 refs/heads/Kemuel@{2}: pull --no-stat -v --progress origin master: Fast-forward

b41c579 HEAD@{2}: pull --no-stat -v --progress origin master: Fast-forward

b41c579 refs/remotes/origin/master@{1}: pull --no-stat -v --progress origin master: fast-forward

e475e27 refs/remotes/origin/Kemuel@{1}: update by push

e475e27 refs/heads/Kemuel@{3}: pull --no-stat -v --progress origin master: Merge made by the 'ort' strategy.

9885c41 refs/remotes/origin/master@{2}: pull --no-stat -v --progress origin master: fast-forward

e475e27 HEAD@{3}: pull --no-stat -v --progress origin master: Merge made by the 'ort' strategy.

5eebb31 refs/remotes/origin/Kemuel@{2}: update by push

5eebb31 refs/heads/Kemuel@{4}: commit: Corrected first sequence diagram v1

5eebb31 HEAD@{4}: commit: Corrected first sequence diagram v1

44d1b83 refs/remotes/origin/Kemuel@{3}: update by push

44d1b83 refs/heads/Kemuel@{5}: commit: Added first sequence diagram

44d1b83 HEAD@{5}: commit: Added first sequence diagram

18e8f5e refs/remotes/origin/Kemuel@{4}: update by push

18e8f5e refs/heads/Kemuel@{6}: commit: Added introduction and framework for Implemented requirements in the Iteration 2 document

18e8f5e HEAD@{6}: commit: Added introduction and framework for Implemented requirements in the Iteration 2 document

5536721 refs/heads/Kemuel@{7}: commit: Added Iteration 1 document to directory

5536721 HEAD@{7}: commit: Added Iteration 1 document to directory

82fcc2c refs/heads/Kemuel@{8}: commit: Corrected Use Case Diagram

82fcc2c HEAD@{8}: commit: Corrected Use Case Diagram
a7a44a0 refs/heads/Kemuel@{9}: commit: Updated Use Case Diagram to current iteration
a7a44a0 HEAD@{9}: commit: Updated Use Case Diagram to current iteration
9dff409 refs/heads/Kemuel@{10}: pull --no-stat -v --progress origin master: Fast-forward
9dff409 refs/remotes/origin/master@{3}: pull --no-stat -v --progress origin master: fast-forward
9dff409 HEAD@{10}: pull --no-stat -v --progress origin master: Fast-forward
cd6eb66 refs/heads/Kemuel@{11}: pull --no-stat -v --progress origin master: Fast-forward
cd6eb66 refs/remotes/origin/master@{4}: pull --no-stat -v --progress origin master: fast-forward
cd6eb66 HEAD@{11}: pull --no-stat -v --progress origin master: Fast-forward
5b0ba70 refs/remotes/origin/Kemuel@{5}: update by push
ac07b6a refs/remotes/origin/Anthony@{0}: fetch origin --recurse-submodules=no --progress --prune:
storing head
253445e refs/remotes/origin/christian@{0}: fetch origin --recurse-submodules=no --progress --prune:
storing head
5b0ba70 refs/heads/Kemuel@{12}: commit: Fixed correctPitch
5b0ba70 HEAD@{12}: commit: Fixed correctPitch
1b1a47a refs/remotes/origin/Kemuel@{6}: update by push
1b1a47a refs/heads/Kemuel@{13}: commit: Added attempts and pitch accuracy to interface
1b1a47a HEAD@{13}: commit: Added attempts and pitch accuracy to interface
1acbff2 refs/heads/Kemuel@{14}: commit: Implemented color changing notes, ENTER and Ctrl+R
command, and Pitch and average Pitch accuracy. Added newAttempt() and resetAttempts()
1acbff2 HEAD@{14}: commit: Implemented color changing notes, ENTER and Ctrl+R command, and
Pitch and average Pitch accuracy. Added newAttempt() and resetAttempts()
606392c refs/heads/Kemuel@{15}: commit: Added getTotalNotes(), setNoteColor(), resetNoteColors()
606392c HEAD@{15}: commit: Added getTotalNotes(), setNoteColor(), resetNoteColors()

Anthony's Logs:

9af52c0 refs/heads/Kemuel@{16}: pull --no-stat -v --progress origin master: Fast-forward
9af52c0 HEAD@{16}: pull --no-stat -v --progress origin master: Fast-forward
9af52c0 refs/remotes/origin/master@{5}: pull --no-stat -v --progress origin master: fast-forward
609deb6 refs/remotes/origin/Kemuel@{7}: update by push
609deb6 refs/heads/Kemuel@{17}: commit: Added setColor to change color of note
609deb6 HEAD@{17}: commit: Added setColor to change color of note

f2ecf9a refs/remotes/origin/Kemuel@{8}: update by push

f2ecf9a refs/heads/Kemuel@{18}: commit: Testing Branch, added test constructor

f2ecf9a HEAD@{18}: commit: Testing Branch, added test constructor

70d245c refs/heads/Kemuel@{19}: branch: Created from HEAD

70d245c HEAD@{19}: checkout: moving from master to Kemuel

70d245c refs/heads/master@{0}: clone: from https://github.com/UNF-CEN4010/G20

70d245c refs/remotes/origin/HEAD@{0}: clone: from https://github.com/UNF-CEN4010/G20

70d245c HEAD@{20}: clone: from https://github.com/UNF-CEN4010/G20

commit 3b576c1687a87c6450d77e9c26f95c3e990367c8

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Sun Mar 29 16:24:29 2026 -0400

Updating Iteration2 pdf with Junit description and UML Diagram

commit 9885c417b19688f00dac484e223ffada7d676475

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Sun Mar 29 05:49:45 2026 -0400

Deleting wrong UML Diagram Iteration 2 file

commit b2d6b14c7ee995001a6c76c5cdc7ee3891f31f3f

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Sun Mar 29 05:48:56 2026 -0400

UML Diagram for Iteration 2 png file

commit 64732fb6320f7a2750e2a95e711653dcd3802462

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Sun Mar 29 05:46:12 2026 -0400

UML Diagram for Iteration 2

commit 39458da9fcb82690036376a5e1ec93d9e987b895

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Sun Mar 29 05:45:22 2026 -0400

Iteration 2 test cases added

commit 00c99b04def87a5444b0a5b0fad74307f1dd449a

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Thu Mar 19 23:21:16 2026 -0400

Extending the Staff to fit more notes

commit b8ab7d571d79f79369c97b2fd8829d0c5a8a44d5

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Thu Mar 19 23:14:46 2026 -0400

Resized window and updated text

commit 1092829b35f26e3fc7c2c2925a957e4c13d68b68

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Thu Mar 19 23:04:42 2026 -0400

Image of keyboard for instruction screen

Christian's Logs:

commit b963299e8030ffe6a1127ff5d95872c7ee42586b

Author: Christian <christianrios246914@gmail.com>

Date: Sun Mar 29 18:35:12 2026 -0400

Swapped lines for a more logical ordering.

commit b41c579699559bac692cb20987d8ed8c555b13d1

Merge: 1be06f1 3b576c1

Author: Christian <christianrios246914@gmail.com>

Date: Sun Mar 29 17:39:33 2026 -0400

Merge branch 'master' of <https://github.com/UNF-CEN4010/G20.git>

commit 1be06f1700f51c71f5436598dfdf0ed2c364ccec

Author: Christian <christianrios246914@gmail.com>

Date: Sun Mar 29 17:36:28 2026 -0400

Changed package names, fixed old test case, simplified new test cases.

commit 3b576c1687a87c6450d77e9c26f95c3e990367c8

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Sun Mar 29 16:22:49 2026 -0400

Updating Iteration2 pdf with Junit description and UML Diagram

commit 80064c30410745d0d61a7c6949ccd533421558ff

Author: Christian <christianrios246914@gmail.com>

Date: Sun Mar 29 15:30:37 2026 -0400

Updated UML class diagram.

commit 80a17eaebb698d563362870b597bcbf8aceee2cf

Merge: 9885c41 e475e27

Author: Levanni <klevannid@gmail.com>

Date: Sun Mar 29 11:56:52 2026 -0400

Merge pull request #11 from UNF-CEN4010/Kemuel

Kemuel

commit e475e27cdd98ebdaac305cef45bcd873055800db

Merge: 5eebb31 9885c41

Author: lxvxnxi <klevannid@gmail.com>

Date: Sun Mar 29 11:53:24 2026 -0400

Merge branch 'master' of <https://github.com/UNF-CEN4010/G20> into Kemuel

commit 9885c417b19688f00dac484e223ffada7d676475

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Sun Mar 29 05:49:45 2026 -0400

Deleting wrong UML Diagram Iteration 2 file

commit b2d6b14c7ee995001a6c76c5cdc7ee3891f31f3f

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Sun Mar 29 05:48:56 2026 -0400

UML Diagram for Iteration 2 png file

commit 64732fb6320f7a2750e2a95e711653dcd3802462

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Sun Mar 29 05:46:12 2026 -0400

UML Diagram for Iteration 2

commit 39458da9fcb82690036376a5e1ec93d9e987b895

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Sun Mar 29 05:45:22 2026 -0400

Iteration 2 test cases added

commit 5eebb31d50751868d94b127aed34c1cdf3b22466

Author: lxvxnxi <klevannid@gmail.com>

Date: Sun Mar 29 04:04:04 2026 -0400

Corrected first sequence diagram v1

commit 44d1b83c3b2e31f1debf692d6601ba21e8a76e14

Author: lxvxnxi <klevannid@gmail.com>

Date: Sun Mar 29 03:45:11 2026 -0400

Added first sequence diagram

commit 18e8f5e9f99f73e24605d40e141dcf4d40aa9ccb

Author: lxvxnxi <klevannid@gmail.com>

Date: Sun Mar 29 02:34:08 2026 -0400

Added introduction and framework for Implemented requirements in the Iteration 2 document

commit 5536721f0e48123523788a1f555c734f9c77f04b

Author: lxvxnxi <klevannid@gmail.com>

Date: Sun Mar 29 02:16:56 2026 -0400

Added Iteration 1 document to directory

Added Iteration 2 blank document to directory

Added Use case diagram to Iteration 2 document

Added Use cases to Iteration 2 document

commit 82fcc2c3448ce3eb862b2875685745f17cc8ae97

Author: lxvxnxi <klevannid@gmail.com>

Date: Sun Mar 29 01:34:23 2026 -0400

Corrected Use Case Diagram

commit a7a44a0cbab26f4f747d9a26382aa4bed072d6f8

Author: lxvxnxi <klevannid@gmail.com>

Date: Sat Mar 28 23:57:05 2026 -0400

Updated Use Case Diagram to current iteration

commit 9dff409fc827df7741b5c03d612224db5e62b612

Merge: 6df0c29 11183b3

Author: christianrios26 <christianrios246914@gmail.com>

Date: Tue Mar 24 09:28:48 2026 -0400

Merge pull request #10 from UNF-CEN4010/christian

Simplified code and improved method names, comments, and encapsulation.

commit 11183b3739cade1a5b6d45250a86954648ce1515

Author: Christian <christianrios246914@gmail.com>

Date: Tue Mar 24 09:27:00 2026 -0400

Simplified code and improved method names, comments, and encapsulation.

commit 6df0c293837935dbf3d0275500f1e5d609db2263

Author: Christian <christianrios246914@gmail.com>

Date: Mon Mar 23 12:56:25 2026 -0400

Changed staff length back from testing value.

commit 174d7bb1ba5439f4b03d2f36eaf9c32c9f672f46

Merge: 94066fd fb54889

Author: christianrios26 <christianrios246914@gmail.com>

Date: Sun Mar 22 20:01:22 2026 -0400

Merge pull request #9 from UNF-CEN4010/christian_refactor

Christian refactor

commit fb5488946d2ca26c6c6dccef126af27c1a7caeb1

Author: Christian <christianrios246914@gmail.com>

Date: Sun Mar 22 19:59:40 2026 -0400

Added comments and improved encapsulation.

commit a3c5359e2fb82c82b44e6ef70bf533bc11675680

Author: Christian <christianrios246914@gmail.com>

Date: Sun Mar 22 18:02:41 2026 -0400

Implemented correct drawing of accidentals within measures.

commit b07f52c61ef493366dc713168d5d44723ce694eb

Author: Christian <christianrios246914@gmail.com>

Date: Sun Mar 22 16:58:00 2026 -0400

Added an additional test for drawing notes on the staff.

commit 1cd6614d852d4c8460e84e1391ed508e530bb1b9

Author: Christian <christianrios246914@gmail.com>

Date: Sun Mar 22 15:45:59 2026 -0400

Added more drawing functionality and heavily refactored.

commit 11df6af21f7c09074412b0e272820aad0c412eb5

Author: Christian <christianrios246914@gmail.com>

Date: Fri Mar 20 20:34:15 2026 -0400

Heavily refactored, created new classes for dynamic note drawing.

commit 471087ed1ccc09c37a3b98fdde233495bda8c4f5

Author: Christian <christianrios246914@gmail.com>

Date: Fri Mar 20 15:24:34 2026 -0400

Repositioned and added text, added ability to go to previous screens.

commit 581a7d89c0c5a0faa327b707e1dd93510ce30e19

Author: Christian <christianrios246914@gmail.com>

Date: Fri Mar 20 13:10:26 2026 -0400

Added new images, refactored image loading.

commit 94066fdf168ca2bf5e29f399e99fd2d6863e3daa

Author: Christian <christianrios246914@gmail.com>

Date: Fri Mar 20 08:28:53 2026 -0400

Increased screen size, repositioned drawings, updated instructions.

commit 2bb5f181d8434d8cedd050b14ea28b3803de1502

Merge: 00c99b0 5f857c4

Author: christianrios26 <christianrios246914@gmail.com>

Date: Fri Mar 20 07:25:00 2026 -0400

Merge pull request #8 from UNF-CEN4010/Kemuel

Kemuel

commit 5f857c4f8adaad16cb8f5cc4298ab7287c3cc59b

Merge: df3430c 00c99b0

Author: christianrios26 <christianrios246914@gmail.com>

Date: Fri Mar 20 07:24:44 2026 -0400

Merge branch 'master' into Kemuel

commit 00c99b04def87a5444b0a5b0fad74307f1dd449a

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Thu Mar 19 23:21:16 2026 -0400

Extending the Staff to fit more notes

commit b8ab7d571d79f79369c97b2fd8829d0c5a8a44d5

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Thu Mar 19 23:14:46 2026 -0400

Resized window and updated text

commit 1092829b35f26e3fc7c2c2925a957e4c13d68b68

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Thu Mar 19 23:04:42 2026 -0400

Image of keyboard for instruction screen

commit df3430caed36071b23eee41c35a3734efac22703

Author: Levanni <klevannid@gmail.com>

Date: Fri Mar 20 01:25:21 2026 +0300

Made accuracy scores only display at the end of an attempt

commit 4f412f401f31a0e4d897ce8af788a4192c552f84

Author: Levanni <klevannid@gmail.com>

Date: Fri Mar 20 00:49:47 2026 +0300

Displays Rhythm Accuracy on screen

commit 7657021be577f102b1bda5a5c307e59ba8fe41bc

Merge: e238f25 bcf8a12

Author: christianrios26 <christianrios246914@gmail.com>

Date: Thu Mar 19 17:14:24 2026 -0400

Merge pull request #7 from UNF-CEN4010/christian

Removed redundant line.

commit bcf8a124b8675c4de33376a868641b7a5a92b91e

Author: Christian <christianrios246914@gmail.com>

Date: Thu Mar 19 17:13:26 2026 -0400

Removed redundant line.

commit e238f2568889bd0d225c96b86740eddfd3e012d2

Author: Christian <christianrios246914@gmail.com>

Date: Thu Mar 19 15:56:06 2026 -0400

Added start screen image and MIDI input detection, shortened class name.

commit c261b167c6271f41f4c02fc5078ab7d58e965721

Author: Christian <christianrios246914@gmail.com>

Date: Mon Mar 16 15:37:22 2026 -0400

Added method for drawing images and refactored the storing of images.

commit 2a9716c68df0b0f6a00d745d1ea207cb805579a2

Author: Christian <christianrios246914@gmail.com>

Date: Sun Mar 8 23:27:18 2026 -0400

Refactored JUnit test cases.

commit 34f8e7364f5b24a4e0d7d30eed3972d74b3fc621

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Sun Mar 8 23:08:59 2026 -0400

Test Cases

commit 4294f1686dd1d25c143a31848bddcc20e2adb4c5

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Sun Mar 8 23:02:11 2026 -0400

Test Cases updated

commit c58e2cf170603d183d7b72335c0f593dd3e953c6

Author: Christian <christianrios246914@gmail.com>

Date: Sun Mar 8 22:27:45 2026 -0400

Some slight code refactoring.

commit dd970cbb81d7f61bea0821ab74dd9f6bf57b2f3b

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Sun Mar 8 19:00:56 2026 -0400

Added additional test cases

commit b6c744583df9ea2e1f12b4d217850daf752a0c26

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Sun Mar 8 17:06:56 2026 -0400

Fixing error

commit 00044463b8933ef03dee08d513c62b26bc6ed5b7

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Sun Mar 8 16:52:07 2026 -0400

adding package

commit 8b6b6c99925a12f8a24fc55b2abc060a90378148

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Sun Mar 8 16:50:16 2026 -0400

Test case for resetAttempts

commit cd6eb66e7709c9e0accd73e7dfedf754278fbc4d

Author: Christian <christianrios246914@gmail.com>

Date: Sun Mar 8 13:40:29 2026 -0400

Finished merging extra screens and note accuracy changes, fixed bug.

commit 6a073d93745b5b434f776ddbeeb86b28446a88412

Merge: c75d4da 51f8839

Author: christianrios26 <christianrios246914@gmail.com>

Date: Sun Mar 8 13:17:55 2026 -0400

Merge pull request #6 from UNF-CEN4010/christian

Merged extra screens with note accuracy functionality.

commit 51f88398194d4b58167b2e5637ffa83e3b22ca23

Merge: 72f8ffc c75d4da

Author: christianrios26 <christianrios246914@gmail.com>

Date: Sun Mar 8 13:17:16 2026 -0400

Merge branch 'master' into christian

commit 72f8ffc110a7dde4ed751d09d7083f04ce589e39

Author: Christian <christianrios246914@gmail.com>

Date: Sun Mar 8 13:03:59 2026 -0400

Updated placeholder instruction screen.

commit c75d4dacc0663380e145a8a64586bd0df89b18b

Merge: 9af52c0 5b0ba70

Author: Levanni <klevannid@gmail.com>

Date: Sun Mar 8 11:56:19 2026 -0400

Merge pull request #5 from UNF-CEN4010/Kemuel

Kemuel

commit 5b0ba7050056d2881999d02b289ac36c4f51b531

Author: lxvxnxi <klevannid@gmail.com>

Date: Sun Mar 8 11:48:47 2026 -0400

Fixed correctPitch

commit b3f81add84787ef0c549952df04fe554c7f573f9

Author: Christian <christianrios246914@gmail.com>

Date: Sun Mar 8 11:27:48 2026 -0400

Added title screen and placeholder instruction screen.

commit 1b1a47aca4ba57e895d737ebf5ee49a9da3820c2

Author: lxvxnxi <klevannid@gmail.com>

Date: Sat Mar 7 22:48:27 2026 -0500

Added attempts and pitch accuracy to interface

commit 1acbff20dd517313667d606ee37682665e9f773b

Author: lxvxnxi <klevannid@gmail.com>

Date: Sat Mar 7 22:46:15 2026 -0500

Implemented color changing notes, ENTER and Ctrl+R command, and Pitch and average Pitch accuracy. Added newAttempt() and resetAttempts()

commit 606392c0caa68df377aae58436a3e61be3602204

Author: lxvxnxi <klevannid@gmail.com>

Date: Sat Mar 7 22:44:25 2026 -0500

Added getTotalNotes(), setNoteColor(), resetNoteColors()

commit 9af52c0a3afb53fe4b349a3cf08127d93c5f46ca

Merge: 4039d5c 253445e

Author: christianrios26 <christianrios246914@gmail.com>

Date: Sat Mar 7 09:58:58 2026 -0500

Merge pull request #4 from UNF-CEN4010/christian

Added detection for keyboard note input and note comparison logic.

commit 253445ec3c6a9fc782a30a2a3fa5d21e4ec0d156

Author: Christian <christianrios246914@gmail.com>

Date: Sat Mar 7 09:55:43 2026 -0500

Added detection for keyboard note input and note comparison logic.

commit 4039d5c3e3b3b1902c20104b0b4c60e31a2e1548

Author: Christian <christianrios246914@gmail.com>

Date: Sat Mar 7 00:56:32 2026 -0500

Fixed note drawing bug.

commit a6f0c597018172729a6861b0d2d076026ccf92ae

Author: christianrios26 <christianrios246914@gmail.com>

Date: Fri Mar 6 22:50:37 2026 -0500

Fixed indentation error.

commit 52bdf4034082b9acc1c767d8faeeb43ef5a1b293

Merge: 1a4f03d e93dff0

Author: christianrios26 <christianrios246914@gmail.com>

Date: Fri Mar 6 22:46:54 2026 -0500

Merge pull request #3 from UNF-CEN4010/christian

Refactored Note class, Staff class, and added basic note drawing.

commit e93dff0c660ddd3f2fbdd8756c0e8cba683daa94

Merge: 284d73e 1a4f03d

Author: christianrios26 <christianrios246914@gmail.com>

Date: Fri Mar 6 22:46:27 2026 -0500

Merge branch 'master' into christian

commit 1a4f03d42a5e8cb69c46d07894484a01cd49b7b4

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Fri Mar 6 22:18:28 2026 -0500

Fixed Typo

commit 284d73e722863f047a3ebdfae6170a465ecddc7c

Author: Christian <christianrios246914@gmail.com>

Date: Fri Mar 6 22:12:43 2026 -0500

Refactored Note class, Staff class, and added basic note drawing.

commit 41c11af61ae980f11eff66de098e481511e1a305

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Fri Mar 6 22:11:41 2026 -0500

Added new font and text being displayed

commit 0adff8604b7561d7f319c08d4e96152427a4569c

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Fri Mar 6 22:05:04 2026 -0500

New font added (Californian FB)

commit 82a3a0d202ae044a8c82d18ece8e67efe2ccae33

Merge: 4fdec80 609deb6

Author: Levanni <klevannid@gmail.com>

Date: Fri Mar 6 11:17:11 2026 -0500

Merge pull request #2 from UNF-CEN4010/Kemuel

Note.Java

commit 4fdec8094ecfea109b457635c6713e5721ec2d9d

Merge: 8e7ff77 704dd6e

Author: christianrios26 <christianrios246914@gmail.com>

Date: Fri Mar 6 09:02:12 2026 -0500

Merge pull request #1 from UNF-CEN4010/chrisdev

Add images, diagrams, and staff drawing

commit 8e7ff773961445fdb6846aabce0ed9b34a5e07d

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Thu Mar 5 22:47:42 2026 -0500

Refactor Staff class with package declaration

commit 704dd6eb2f8008e12b0a177d061e732c0087b7dc

Author: Christian <christianrios246914@gmail.com>

Date: Thu Mar 5 22:37:42 2026 -0500

Added keyboard input detection (test for Enter key input).

commit 609deb60739f48a5ad37e91bb866129c06bca75f

Author: lxvxnxi <klevannid@gmail.com>

Date: Thu Mar 5 22:17:59 2026 -0500

Added setColor to change color of note

commit ec2fe8b78652d1cb0395fed800280800f9be19d3

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Thu Mar 5 22:12:21 2026 -0500

Add files via upload

commit f2ecf9a7de9f8f6f959da2363d423e70e94576da

Author: lxvxnx <klevannid@gmail.com>

Date: Thu Mar 5 22:06:43 2026 -0500

Testing Branch, added test constructor

commit bb92cbab9c1308b93c5e6386873cc7e76e5723dd

Author: Christian <christianrios246914@gmail.com>

Date: Thu Mar 5 22:01:06 2026 -0500

Added initial UML and use case diagrams.

commit ac07b6a5893ee12e9685cf07acbc4d9f962c9a28

Author: ThePotatoSalad <anoschese25@gmail.com>

Date: Thu Mar 5 21:59:22 2026 -0500

Add files via upload

commit 9db75f5bc57370b294c8c6c81dbaf507643c77a9

Author: Christian <christianrios246914@gmail.com>

Date: Thu Mar 5 21:55:07 2026 -0500

Added staff drawing with clef and time signature.

commit 79f67f85944b3d243cc4efd705a7ca82638f96ba

Author: Christian <christianrios246914@gmail.com>

Date: Thu Mar 5 21:38:40 2026 -0500

Added image files.

commit 70d245c9f4b45d39cf8d529d05ae363d127c438b

Author: Christian <christianrios246914@gmail.com>

Date: Sun Mar 1 13:31:12 2026 -0500

Added additional files in the Eclipse project folder.

commit a7ba8b2e21b44b8aadd2a738122647b5c925b9f2

Author: Christian <christianrios246914@gmail.com>

Date: Sun Mar 1 13:25:42 2026 -0500

Initial project commit

b963299 refs/remotes/origin/master@{0}: push: forced-update

b963299 refs/heads/master@{0}: commit: Swapped lines for a more logical ordering.

b963299 HEAD@{0}: commit: Swapped lines for a more logical ordering.

b41c579 refs/remotes/origin/master@{1}: push: forced-update

b41c579 refs/heads/master@{1}: merge branch 'master' of <https://github.com/UNF-CEN4010/G20.git>: Merge made by recursive.

3b576c1 refs/remotes/origin/master@{2}: fetch: forced-update

b41c579 HEAD@{1}: merge branch 'master' of <https://github.com/UNF-CEN4010/G20.git>: Merge made by recursive.

1be06f1 HEAD@{2}: checkout: moving from christian to master

1be06f1 refs/heads/christian@{0}: branch: Created from branch refs/heads/master

1be06f1 HEAD@{3}: checkout: moving from master to christian

1be06f1 refs/heads/master@{2}: commit: Changed package names, fixed old test case, simplified new test cases.

1be06f1 HEAD@{4}: commit: Changed package names, fixed old test case, simplified new test cases.

80064c3 refs/remotes/origin/master@{3}: push: forced-update

80064c3 refs/heads/master@{3}: commit: Updated UML class diagram.

80064c3 HEAD@{5}: commit: Updated UML class diagram.

80a17ea refs/heads/master@{4}: merge branch 'master' of <https://github.com/UNF-CEN4010/G20.git>: Fast-forward

e475e27 refs/remotes/origin/Kemuel@{0}: fetch: forced-update
80a17ea refs/remotes/origin/master@{4}: fetch: forced-update
80a17ea HEAD@{6}: merge branch 'master' of https://github.com/UNF-CEN4010/G20.git: Fast-forward
9dff409 HEAD@{7}: checkout: moving from christian to master
11183b3 HEAD@{8}: checkout: moving from master to christian
9dff409 refs/heads/master@{5}: merge branch 'master' of https://github.com/UNF-CEN4010/G20.git: Fast-forward
9dff409 refs/remotes/origin/master@{5}: fetch: forced-update
9dff409 HEAD@{9}: merge branch 'master' of https://github.com/UNF-CEN4010/G20.git: Fast-forward
6df0c29 HEAD@{10}: checkout: moving from christian to master
11183b3 refs/remotes/origin/christian@{0}: push: forced-update
11183b3 HEAD@{11}: commit: Simplified code and improved method names, comments, and encapsulation.
174d7bb HEAD@{12}: checkout: moving from master to christian
6df0c29 refs/remotes/origin/master@{6}: push: forced-update
6df0c29 refs/heads/master@{6}: commit: Changed staff length back from testing value.
6df0c29 HEAD@{13}: commit: Changed staff length back from testing value.
174d7bb HEAD@{14}: checkout: moving from christian to master
174d7bb HEAD@{15}: checkout: moving from master to christian
174d7bb HEAD@{16}: checkout: moving from christian to master
174d7bb HEAD@{17}: checkout: moving from master to christian
174d7bb refs/heads/master@{7}: merge branch 'master' of https://github.com/UNF-CEN4010/G20.git: Fast-forward
174d7bb refs/remotes/origin/master@{7}: fetch: forced-update
174d7bb HEAD@{18}: merge branch 'master' of https://github.com/UNF-CEN4010/G20.git: Fast-forward
94066fd HEAD@{19}: checkout: moving from christian_refactor to master
fb54889 refs/remotes/origin/christian_refactor@{0}: push: forced-update
fb54889 HEAD@{20}: commit: Added comments and improved encapsulation.
a3c5359 refs/remotes/origin/christian_refactor@{1}: push: forced-update
a3c5359 HEAD@{21}: commit: Implemented correct drawing of accidentals within measures.
b07f52c refs/remotes/origin/christian_refactor@{2}: push: forced-update

b07f52c HEAD@{22}: commit: Added an additional test for drawing notes on the staff.

1cd6614 refs/remotes/origin/christian_refactor@{3}: push: forced-update

1cd6614 HEAD@{23}: commit: Added more drawing functionality and heavily refactored.

11df6af refs/remotes/origin/christian_refactor@{4}: push: created

11df6af HEAD@{24}: commit: Heavily refactored, created new classes for dynamic note drawing.

471087e HEAD@{25}: checkout: moving from christian to christian_refactor

471087e refs/remotes/origin/christian@{1}: push: forced-update

471087e HEAD@{26}: commit: Repositioned and added text, added ability to go to previous screens.

581a7d8 refs/remotes/origin/christian@{2}: push: forced-update

581a7d8 HEAD@{27}: commit: Added new images, refactored image loading.

94066fd HEAD@{28}: checkout: moving from master to christian

94066fd refs/remotes/origin/master@{8}: push: forced-update

94066fd refs/heads/master@{8}: commit: Increased screen size, repositioned drawings, updated instructions.

94066fd HEAD@{29}: commit: Increased screen size, repositioned drawings, updated instructions.

2bb5f18 refs/heads/master@{9}: merge branch 'master' of https://github.com/UNF-CEN4010/G20.git: Fast-forward

2bb5f18 HEAD@{30}: merge branch 'master' of https://github.com/UNF-CEN4010/G20.git: Fast-forward

00c99b0 HEAD@{31}: checkout: moving from Kemuel to master

5f857c4 refs/heads/Kemuel@{0}: merge branch 'Kemuel' of https://github.com/UNF-CEN4010/G20.git: Fast-forward

5f857c4 refs/remotes/origin/Kemuel@{1}: fetch: forced-update

2bb5f18 refs/remotes/origin/master@{9}: fetch: forced-update

5f857c4 HEAD@{32}: merge branch 'Kemuel' of https://github.com/UNF-CEN4010/G20.git: Fast-forward

df3430c refs/heads/Kemuel@{1}: branch: Created from branch refs/remotes/origin/Kemuel

df3430c HEAD@{33}: checkout: moving from master to Kemuel

00c99b0 refs/heads/master@{10}: merge branch 'master' of https://github.com/UNF-CEN4010/G20.git: Fast-forward

df3430c refs/remotes/origin/Kemuel@{2}: fetch: forced-update

00c99b0 refs/remotes/origin/master@{10}: fetch: forced-update

00c99b0 HEAD@{34}: merge branch 'master' of https://github.com/UNF-CEN4010/G20.git: Fast-forward

e238f25 HEAD@{35}: checkout: moving from Anthony to master

4294f16 HEAD@{36}: checkout: moving from christian to Anthony

3fb7929 HEAD@{37}: commit: Updated text visibility.

bcf8a12 refs/remotes/origin/christian@{3}: push: forced-update

bcf8a12 HEAD@{38}: commit: Removed redundant line.

e238f25 HEAD@{39}: checkout: moving from master to christian

e238f25 refs/remotes/origin/master@{11}: push: forced-update

e238f25 refs/heads/master@{11}: commit: Added start screen image and MIDI input detection, shortened class name.

e238f25 HEAD@{40}: commit: Added start screen image and MIDI input detection, shortened class name.

c261b16 refs/remotes/origin/master@{12}: push: forced-update

c261b16 refs/heads/master@{12}: commit: Added method for drawing images and refactored the storing of images.

c261b16 HEAD@{41}: commit: Added method for drawing images and refactored the storing of images.

2a9716c refs/remotes/origin/master@{13}: push: forced-update

2a9716c refs/heads/master@{13}: commit: Refactored JUnit test cases.

2a9716c HEAD@{42}: commit: Refactored JUnit test cases.

34f8e73 refs/heads/master@{14}: merge branch 'master' of <https://github.com/UNF-CEN4010/G20.git>: Fast-forward

4294f16 refs/remotes/origin/Anthony@{0}: fetch: forced-update

34f8e73 refs/remotes/origin/master@{14}: fetch: forced-update

34f8e73 HEAD@{43}: merge branch 'master' of <https://github.com/UNF-CEN4010/G20.git>: Fast-forward

c58e2cf refs/remotes/origin/master@{15}: push: forced-update

c58e2cf refs/heads/master@{15}: commit: Some slight code refactoring.

c58e2cf HEAD@{44}: commit: Some slight code refactoring.

cd6eb66 refs/remotes/origin/master@{16}: push: forced-update

cd6eb66 refs/heads/master@{16}: commit: Finished merging extra screens and note accuracy changes, fixed bug.

cd6eb66 HEAD@{45}: commit: Finished merging extra screens and note accuracy changes, fixed bug.

6a073d9 refs/heads/master@{17}: merge branch 'master' of <https://github.com/UNF-CEN4010/G20.git>: Fast-forward

5b0ba70 refs/remotes/origin/Kemuel@{3}: fetch: forced-update

6a073d9 refs/remotes/origin/master@{17}: fetch: forced-update

6a073d9 HEAD@{46}: merge branch 'master' of https://github.com/UNF-CEN4010/G20.git: Fast-forward
4039d5c HEAD@{47}: checkout: moving from christian to master
72f8ffc refs/remotes/origin/christian@{4}: push: forced-update
72f8ffc HEAD@{48}: commit: Updated placeholder instruction screen.
b3f81ad HEAD@{49}: checkout: moving from master to christian
4039d5c HEAD@{50}: checkout: moving from christian to master
b3f81ad HEAD@{51}: checkout: moving from Kemuel to christian
1b1a47a HEAD@{52}: checkout: moving from christian to Kemuel
b3f81ad HEAD@{53}: checkout: moving from Kemuel to christian
1b1a47a refs/remotes/origin/Kemuel@{4}: fetch: forced-update
9af52c0 refs/remotes/origin/master@{18}: fetch: forced-update
1b1a47a HEAD@{54}: merge branch 'Kemuel' of https://github.com/UNF-CEN4010/G20.git: Fast-forward
609deb6 HEAD@{55}: checkout: moving from christian to Kemuel
b3f81ad HEAD@{56}: checkout: moving from Kemuel to christian
609deb6 HEAD@{57}: checkout: moving from christian to Kemuel
b3f81ad HEAD@{58}: commit: Added title screen and placeholder instruction screen.
253445e refs/remotes/origin/christian@{5}: push: forced-update
253445e HEAD@{59}: commit: Added detection for keyboard note input and note comparison logic.
4039d5c HEAD@{60}: checkout: moving from master to christian
4039d5c refs/remotes/origin/master@{19}: push: forced-update
4039d5c refs/heads/master@{18}: commit: Fixed note drawing bug.
4039d5c HEAD@{61}: commit: Fixed note drawing bug.
a6f0c59 refs/heads/master@{19}: merge branch 'master' of https://github.com/UNF-CEN4010/G20.git: Fast-forward
a6f0c59 refs/remotes/origin/master@{20}: fetch: forced-update
a6f0c59 HEAD@{62}: merge branch 'master' of https://github.com/UNF-CEN4010/G20.git: Fast-forward
82a3a0d HEAD@{63}: checkout: moving from christian to master
284d73e refs/remotes/origin/christian@{6}: push: created
284d73e HEAD@{64}: commit: Refactored Note class, Staff class, and added basic note drawing.
82a3a0d HEAD@{65}: checkout: moving from master to christian

82a3a0d refs/heads/master@{20}: merge branch 'master' of https://github.com/UNF-CEN4010/G20.git: Fast-forward

82a3a0d refs/remotes/origin/master@{21}: fetch: forced-update

82a3a0d HEAD@{66}: merge branch 'master' of https://github.com/UNF-CEN4010/G20.git: Fast-forward

4fdec80 HEAD@{67}: checkout: moving from chrisdev to master

704dd6e HEAD@{68}: checkout: moving from master to chrisdev

4fdec80 refs/heads/master@{21}: merge branch 'master' of https://github.com/UNF-CEN4010/G20.git: Fast-forward

4fdec80 HEAD@{69}: merge branch 'master' of https://github.com/UNF-CEN4010/G20.git: Fast-forward

70d245c HEAD@{70}: checkout: moving from chrisdev to master

ac07b6a refs/remotes/origin/Anthony@{1}: fetch: created

609deb6 refs/remotes/origin/Kemuel@{5}: fetch: created

4fdec80 refs/remotes/origin/master@{22}: fetch: forced-update

704dd6e HEAD@{71}: commit: Added keyboard input detection (test for Enter key input).

bb92cba HEAD@{72}: commit: Added initial UML and use case diagrams.

9db75f5 HEAD@{73}: commit: Added staff drawing with clef and time signature.

79f67f8 HEAD@{74}: commit: Added image files.

70d245c HEAD@{75}: checkout: moving from master to chrisdev

70d245c refs/remotes/origin/master@{23}: push: forced-update

70d245c refs/heads/master@{22}: commit: Added additional files in the Eclipse project folder.

70d245c HEAD@{76}: commit: Added additional files in the Eclipse project folder.

a7ba8b2 refs/remotes/origin/master@{24}: push: created

a7ba8b2 refs/heads/master@{23}: commit (initial): Initial project commit

a7ba8b2 HEAD@{77}: commit (initial): Initial project commit